

80 LL EXTI Generic Driver

80.1 EXTI Firmware driver registers structures

80.1.1 LL_EXTI_InitTypeDef

LL_EXTI_InitTypeDef is defined in the `stm32f4xx_ll_exti.h`

Data Fields

- *uint32_t Line_0_31*
- *FunctionalState LineCommand*
- *uint8_t Mode*
- *uint8_t Trigger*

Field Documentation

- *uint32_t LL_EXTI_InitTypeDef::Line_0_31*
Specifies the EXTI lines to be enabled or disabled for Lines in range 0 to 31 This parameter can be any combination of *EXTI_LL_EC_LINE*
- *FunctionalState LL_EXTI_InitTypeDef::LineCommand*
Specifies the new state of the selected EXTI lines. This parameter can be set either to ENABLE or DISABLE
- *uint8_t LL_EXTI_InitTypeDef::Mode*
Specifies the mode for the EXTI lines. This parameter can be a value of *EXTI_LL_EC_MODE*.
- *uint8_t LL_EXTI_InitTypeDef::Trigger*
Specifies the trigger signal active edge for the EXTI lines. This parameter can be a value of *EXTI_LL_EC_TRIGGER*.

80.2 EXTI Firmware driver API description

The following section lists the various functions of the EXTI library.

80.2.1 Detailed description of functions

LL_EXTI_EnableIT_0_31

Function name

`__STATIC_INLINE void LL_EXTI_EnableIT_0_31 (uint32_t ExtiLine)`

Function description

Enable ExtiLine Interrupt request for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **None:**

Notes

- The reset value for the direct or internal lines (see RM) is set to 1 in order to enable the interrupt by default. Bits are set automatically at Power on.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- IMR IMx LL_EXTI_EnableIT_0_31

LL_EXTI_DisableIT_0_31

Function name

__STATIC_INLINE void LL_EXTI_DisableIT_0_31 (uint32_t ExtiLine)

Function description

Disable ExtiLine Interrupt request for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **None:**

Notes

- The reset value for the direct or internal lines (see RM) is set to 1 in order to enable the interrupt by default. Bits are set automatically at Power on.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- IMR IMx LL_EXTI_DisableIT_0_31

LL_EXTI_IsEnabledIT_0_31

Function name

__STATIC_INLINE uint32_t LL_EXTI_IsEnabledIT_0_31 (uint32_t ExtiLine)

Function description

Indicate if ExtiLine Interrupt request is enabled for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **State:** of bit (1 or 0).

Notes

- The reset value for the direct or internal lines (see RM) is set to 1 in order to enable the interrupt by default. Bits are set automatically at Power on.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- IMR IMx LL_EXTI_IsEnabledIT_0_31

LL_EXTI_EnableEvent_0_31

Function name

__STATIC_INLINE void LL_EXTI_EnableEvent_0_31 (uint32_t ExtiLine)

Function description

Enable ExtiLine Event request for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **None:**

Notes

- (*) : Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- EMR EMx LL_EXTI_EnableEvent_0_31

LL_EXTI_DisableEvent_0_31

Function name

__STATIC_INLINE void LL_EXTI_DisableEvent_0_31 (uint32_t ExtiLine)

Function description

Disable ExtiLine Event request for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **None:**

Notes

- (*) : Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- EMR EMx LL_EXTI_DisableEvent_0_31

LL_EXTI_IsEnabledEvent_0_31

Function name

`__STATIC_INLINE uint32_t LL_EXTI_IsEnabledEvent_0_31 (uint32_t ExtiLine)`

Function description

Indicate if ExtiLine Event request is enabled for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be one of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_17
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22
 - LL_EXTI_LINE_23(*)
 - LL_EXTI_LINE_ALL_0_31

Return values

- **State:** of bit (1 or 0).

Notes

- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- EMR EMx LL_EXTI_IsEnabledEvent_0_31

LL_EXTI_EnableRisingTrig_0_31

Function name

`__STATIC_INLINE void LL_EXTI_EnableRisingTrig_0_31 (uint32_t ExtiLine)`

Function description

Enable ExtiLine Rising Edge Trigger for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- The configurable wakeup lines are edge-triggered. No glitch must be generated on these lines. If a rising edge on a configurable interrupt line occurs during a write operation in the EXTI_RTSR register, the pending bit is not set. Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- RTSR RTx LL_EXTI_EnableRisingTrig_0_31

LL_EXTI_DisableRisingTrig_0_31

Function name

__STATIC_INLINE void LL_EXTI_DisableRisingTrig_0_31 (uint32_t ExtiLine)

Function description

Disable ExtiLine Rising Edge Trigger for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- The configurable wakeup lines are edge-triggered. No glitch must be generated on these lines. If a rising edge on a configurable interrupt line occurs during a write operation in the EXTI_RTSR register, the pending bit is not set. Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- RTSR RTx LL_EXTI_DisableRisingTrig_0_31

LL_EXTI_IsEnabledRisingTrig_0_31

Function name

__STATIC_INLINE uint32_t LL_EXTI_IsEnabledRisingTrig_0_31 (uint32_t ExtiLine)

Function description

Check if rising edge trigger is enabled for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **State:** of bit (1 or 0).

Notes

- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- RTSR RTx LL_EXTI_IsEnabledRisingTrig_0_31

LL_EXTI_EnableFallingTrig_0_31

Function name

__STATIC_INLINE void LL_EXTI_EnableFallingTrig_0_31 (uint32_t ExtiLine)

Function description

Enable ExtiLine Falling Edge Trigger for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- The configurable wakeup lines are edge-triggered. No glitch must be generated on these lines. If a falling edge on a configurable interrupt line occurs during a write operation in the EXTI_FTSR register, the pending bit is not set. Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- FTSR FTx LL_EXTI_EnableFallingTrig_0_31

LL_EXTI_DisableFallingTrig_0_31

Function name

__STATIC_INLINE void LL_EXTI_DisableFallingTrig_0_31 (uint32_t ExtiLine)

Function description

Disable ExtiLine Falling Edge Trigger for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- The configurable wakeup lines are edge-triggered. No glitch must be generated on these lines. If a Falling edge on a configurable interrupt line occurs during a write operation in the EXTI_FTSR register, the pending bit is not set. Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- FTSR FTx LL_EXTI_DisableFallingTrig_0_31

LL_EXTI_IsEnabledFallingTrig_0_31

Function name

__STATIC_INLINE uint32_t LL_EXTI_IsEnabledFallingTrig_0_31 (uint32_t ExtiLine)

Function description

Check if falling edge trigger is enabled for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **State:** of bit (1 or 0).

Notes

- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- FTSR FTx LL_EXTI_IsEnabledFallingTrig_0_31

LL_EXTI_GenerateSWI_0_31

Function name

__STATIC_INLINE void LL_EXTI_GenerateSWI_0_31 (uint32_t ExtiLine)

Function description

Generate a software Interrupt Event for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- If the interrupt is enabled on this line in the EXTI_IMR, writing a 1 to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation. This bit is cleared by clearing the corresponding bit in the EXTI_PR register (by writing a 1 into the bit)
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- SWIER SWIx LL_EXTI_GenerateSWI_0_31

LL_EXTI_IsActiveFlag_0_31

Function name

__STATIC_INLINE uint32_t LL_EXTI_IsActiveFlag_0_31 (uint32_t ExtiLine)

Function description

Check if the ExtLine Flag is set or not for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **State:** of bit (1 or 0).

Notes

- This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a 1 to the bit.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- PR PIFx LL_EXTI_IsActiveFlag_0_31

LL_EXTI_ReadFlag_0_31

Function name

`__STATIC_INLINE uint32_t LL_EXTI_ReadFlag_0_31 (uint32_t ExtiLine)`

Function description

Read ExtLine Combination Flag for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **@note:** This bit is set when the selected edge event arrives on the interrupt

Notes

- This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a 1 to the bit.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- PR PIFx LL_EXTI_ReadFlag_0_31

LL_EXTI_ClearFlag_0_31

Function name

__STATIC_INLINE void LL_EXTI_ClearFlag_0_31 (uint32_t ExtiLine)

Function description

Clear ExtLine Flags for Lines in range 0 to 31.

Parameters

- **ExtiLine:** This parameter can be a combination of the following values:
 - LL_EXTI_LINE_0
 - LL_EXTI_LINE_1
 - LL_EXTI_LINE_2
 - LL_EXTI_LINE_3
 - LL_EXTI_LINE_4
 - LL_EXTI_LINE_5
 - LL_EXTI_LINE_6
 - LL_EXTI_LINE_7
 - LL_EXTI_LINE_8
 - LL_EXTI_LINE_9
 - LL_EXTI_LINE_10
 - LL_EXTI_LINE_11
 - LL_EXTI_LINE_12
 - LL_EXTI_LINE_13
 - LL_EXTI_LINE_14
 - LL_EXTI_LINE_15
 - LL_EXTI_LINE_16
 - LL_EXTI_LINE_18
 - LL_EXTI_LINE_19(*)
 - LL_EXTI_LINE_20(*)
 - LL_EXTI_LINE_21
 - LL_EXTI_LINE_22

Return values

- **None:**

Notes

- This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a 1 to the bit.
- (*): Available in some devices
- Please check each device line mapping for EXTI Line availability

Reference Manual to LL API cross reference:

- PR PIFx LL_EXTI_ClearFlag_0_31

LL_EXTI_Init

Function name

uint32_t LL_EXTI_Init (LL_EXTI_InitTypeDef * EXTI_InitStruct)

Function description

Initialize the EXTI registers according to the specified parameters in EXTI_InitStruct.

Parameters

- **EXTI_InitStruct:** pointer to a LL_EXTI_InitTypeDef structure.

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: EXTI registers are initialized
 - ERROR: not applicable

LL_EXTI_DeInit

Function name

uint32_t LL_EXTI_DeInit (void)

Function description

De-initialize the EXTI registers to their default reset values.

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: EXTI registers are de-initialized
 - ERROR: not applicable

LL_EXTI_StructInit

Function name

void LL_EXTI_StructInit (LL_EXTI_InitTypeDef * EXTI_InitStruct)

Function description

Set each LL_EXTI_InitTypeDef field to default value.

Parameters

- **EXTI_InitStruct:** Pointer to a LL_EXTI_InitTypeDef structure.

Return values

- **None:**

80.3 EXTI Firmware driver defines

The following section lists the various define and macros of the module.

80.3.1 EXTI

EXTI
LINE

LL_EXTI_LINE_0

Extended line 0

LL_EXTI_LINE_1

Extended line 1

LL_EXTI_LINE_2

Extended line 2

LL_EXTI_LINE_3

Extended line 3

LL_EXTI_LINE_4

Extended line 4

LL_EXTI_LINE_5

Extended line 5

LL_EXTI_LINE_6

Extended line 6

LL_EXTI_LINE_7

Extended line 7

LL_EXTI_LINE_8

Extended line 8

LL_EXTI_LINE_9

Extended line 9

LL_EXTI_LINE_10

Extended line 10

LL_EXTI_LINE_11

Extended line 11

LL_EXTI_LINE_12

Extended line 12

LL_EXTI_LINE_13

Extended line 13

LL_EXTI_LINE_14

Extended line 14

LL_EXTI_LINE_15

Extended line 15

LL_EXTI_LINE_16

Extended line 16

LL_EXTI_LINE_17

Extended line 17

LL_EXTI_LINE_18

Extended line 18

LL_EXTI_LINE_19

Extended line 19

LL_EXTI_LINE_20

Extended line 20

LL_EXTI_LINE_21

Extended line 21

LL_EXTI_LINE_22

Extended line 22

LL_EXTI_LINE_ALL_0_31

All Extended line not reserved

LL_EXTI_LINE_ALL

All Extended line

LL_EXTI_LINE_NONE

None Extended line

Mode

LL_EXTI_MODE_IT

Interrupt Mode

LL_EXTI_MODE_EVENT

Event Mode

LL_EXTI_MODE_IT_EVENT

Interrupt & Event Mode

Edge Trigger**LL_EXTI_TRIGGER_NONE**

No Trigger Mode

LL_EXTI_TRIGGER_RISING

Trigger Rising Mode

LL_EXTI_TRIGGER_FALLING

Trigger Falling Mode

LL_EXTI_TRIGGER_RISING_FALLING

Trigger Rising & Falling Mode

Common Write and read registers Macros**LL_EXTI_WriteReg****Description:**

- Write a value in EXTI register.

Parameters:

- `__REG__`: Register to be written
- `__VALUE__`: Value to be written in the register

Return value:

- None

LL_EXTI_ReadReg**Description:**

- Read a value in EXTI register.

Parameters:

- `__REG__`: Register to be read

Return value:

- Register: value